



Global COVID-19 Insights: Death, Vaccination, and Policy Impacts

➤ Objective:

Perform a comprehensive end-to-end SQL data analysis project that covers:

- Data Cleaning & Transformation
- Aggregations & KPIs
- Joins between datasets
- Time-series analysis
- Correlation between vaccination and death trends
- Country and continent-level drill-down
- Window functions, subqueries, CTEs, case statements, etc.

➤ Project Scope: Combined Analysis

Area	Skills Involved
Data Cleaning	NULLs, date formatting, derived columns
Data Exploration	Aggregation, grouping, filtering
Data Transformation	Joins, CTEs, calculated fields
Window Functions	Rank countries, % change, cumulative sum
Case Statements	Categorize based on vaccination levels
Joins	Deaths + Vaccination
Visualization Prer	Query output to CSV or BI dashboards

➤ Suggested Final Table Name

covid_full_data : (Merge of covid_deaths and covid_vaccination using location and date)



➤ Suggested Schema After Cleaning

Column	Description
location	Country name
continent	Continent
date	Daily Record
population	Population of the Country
total_cases	Cumulative Cases
new_cases	Daily New Cases
total_deaths	Cumulative deaths
new_deaths	Daily New Deaths
total_vaccinations	Cumulative Vaccination Doses
people_vaccinated	People With at Least One Dose
people_Fully_vaccinated	fully vaccinated people
new_vaccinations	Daily Vaccinations
death_rate	Derived: deaths / cases
vaccination_rate	Derived: vaccinated / population
fully_vaccinated_pct	Derived: fully vaccinated / population
case_rate	Derived: total_cases / population

➤ Key area Analysis

1.Global KPIs

- Total cases, deaths, vaccinations globally
- Global death rate
- Global vaccination rate

2.Country-Level Analysis

- Top 10 countries by: Total deaths, Total cases per capita, Vaccination Rate
- Worst-hit countries by death rate
- Best-performing countries in vaccination vs. case count

3. Time-Series Analysis

- Daily new cases, deaths, vaccinations
- Moving average of cases/deaths (7-day, 14-day)
- Vaccination vs. death rate over time
- Case surge detection using conditional logic

4. Continent Comparision

- Cases and deaths per million by continent
- Total vaccinations per continent
- Vaccination success rate (fully vaccinated vs total)

5. Correlation Analysis

- Vaccination rate vs death rate
- Cases vs deaths (time lagged)
- Impact of reaching 50% vaccination milestone

6. Advanced SQL Features

- **Window Functions:** Cumulative sum, rank, moving average
- **CTEs:** Cleaned datasets, daily changes, categorized country list
- **Joins:** Merge deaths and vaccinations using location + date
- **CASE Statements:** Bucket countries (e.g., Low, Medium, High Vaccination)
- **Derived Columns:** Death rate, vaccination rate, etc.

➤ Final Deliverables

Components	Description
SQL Script	For table creation, cleaning, analysis
KPI Dashboard	In Power BI or Tableau (optional)
Analytical Report	Insight summary + policy suggestions
Exportable CSVs	Time-series trends, top countries